TOPIC 05

Sampling Distributions

Why a statistic is itself a random variable, and the exact formulas for how the sample mean and proportion behave

WHAT YOU WILL LEARN

- Explain why a sample statistic varies and what its sampling distribution is
- Keep the population distribution, sample distribution and sampling distribution apart
- State and use the mean and standard deviation of the sampling distribution of $\bar{x}$
- Describe the sampling distribution of a sample proportion $\hat{p}$
- Know when the finite population correction is needed

1. A statistic is a random variable

In Topic 01 we saw that two samples from the same population give different means. Before we draw a sample, its mean $\bar{x}$ is **unknown and uncertain**, so $\bar{x}$ is itself a **random variable**. Like any random variable it has a distribution, called the **sampling distribution**.

DEFINITION **Sampling distribution**

The sampling distribution of a statistic is the distribution of the values that statistic would take over **all possible random samples of the same size n** from the population.

Imagine this thought experiment: draw a sample of size n, compute $\bar{x}$, put the sample back, and repeat thousands of times. A histogram of all those $\bar{x}$ values is the sampling distribution of the mean. In real life we draw only **one** sample; the sampling distribution tells us how far that one result is likely to be from the truth.

2. Three distributions, easily confused

	Population distribution	Sample (data) distribution	Sampling distribution of $\bar{x}$
What is plotted	Every value in the population	The n values in your one sample	The $\bar{x}$ from many samples of size n
Centre	μ	$\bar{x}$	μ
Spread	σ	s	$\sigma/\sqrt{n}$ (standard error)
Known?	Usually not	Yes, it is your data	Derived by theory
Changes with n?	No	Gets closer to the population shape	Gets narrower and more normal

KEY IDEA **The most important idea in inference**

Inference is built on the **third** column. We reason about a single $\bar{x}$ by asking where it sits in its sampling distribution.

3. Seeing it: a simulation

Our population has 100,000 values and is strongly right-skewed, with μ = 60.5 and σ = 50.5. For each sample size we drew 4,000 random samples and recorded each x̄.

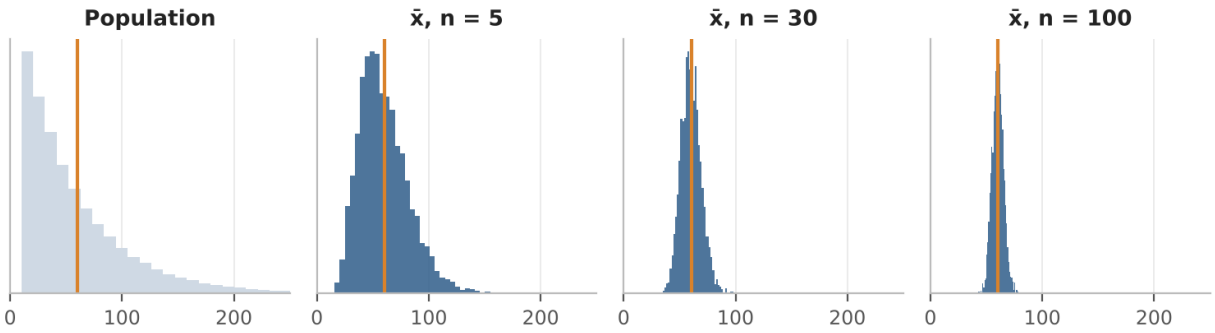


Figure 1. The skewed population (left) and the sampling distribution of x̄ for n = 5, 30, 100. The orange line is μ. Every sampling distribution centres on μ and narrows as n grows.

Sample size n	Mean of the 4,000 x̄	SD of the 4,000 x̄	Theory: σ/√n
5	59.98	22.25	22.58
30	60.35	9.07	9.22
100	60.31	5.07	5.05

The simulated numbers match the theory almost exactly. That theory is the next section.

4. The sampling distribution of the sample mean

DEFINITION **Result (any population with mean μ and standard deviation σ)**

- **Centre:** E(x̄) = μ. The sample mean is **unbiased**: on average it hits the target.
- **Spread:** the standard deviation of x̄ is σ/√n. It is called the **standard error** (Topic 07).
- **Shape:** if the population is normal, x̄ is exactly normal for any n. Otherwise it becomes approximately normal as n grows (Topic 06).

$$E(\bar{x}) = \mu, \qquad \mathrm{Var}(\bar{x}) = \frac{\sigma^2}{n}, \qquad \mathrm{SD}(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

Where does σ/√n come from?

Write x̄ = (X₁ + X₂ + ... + Xₙ)/n with independent observations, each with variance σ². The variance of a sum of independent variables is the sum of the variances (Topic 02), so Var(X₁ + ... + Xₙ) = nσ². Dividing by n divides the variance by n², giving

$$\mathrm{Var}(\bar{x}) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

Averaging cancels out some of the random ups and downs. Larger n means more cancellation and a tighter x̄. Note the **square root**: to halve the spread you need four times the data.

When the population is normal

$$X \sim N(\mu, \sigma^2) \implies \bar{x} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \implies Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

WORKED EXAMPLE **Heights of adults**

Adult heights are N(170, 8²) cm. A random sample of n = 25 people is taken. The standard deviation of $\bar{x}$ is 8/√25 = **1.6 cm**. What is P($\bar{x}$ > 172)? z = (172 − 170)/1.6 = 1.25, so the probability is **0.1056**. Compare an *individual* taller than 172 cm: z = 0.25, probability 0.4013. Averages vary much less than individuals.

5. The sampling distribution of a proportion

When the data is yes/no, the statistic of interest is the **sample proportion** $\hat{p}$ = (number of successes)/n. Each observation is a Bernoulli trial with mean p and variance p(1 − p), so the same logic gives:

$$E(\hat{p}) = p, \qquad \mathrm{SD}(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

$\hat{p}$ is approximately normal when both **np ≥ 10** and **n(1 − p) ≥ 10** (the 'success-failure condition').

WORKED EXAMPLE **Support for a policy**

Suppose 60% of a population supports a policy (p = 0.6) and we poll n = 200. Conditions: np = 120 and n(1 − p) = 80, both at least 10. SD($\hat{p}$) = √(0.6 × 0.4/200) = **0.0346**. P($\hat{p}$ < 0.55) = P(Z < (0.55 − 0.60)/0.0346) = P(Z < -1.44) = **0.0745**. So a poll showing under 55% from a population with true support 60% is possible, but only about 7% of the time.

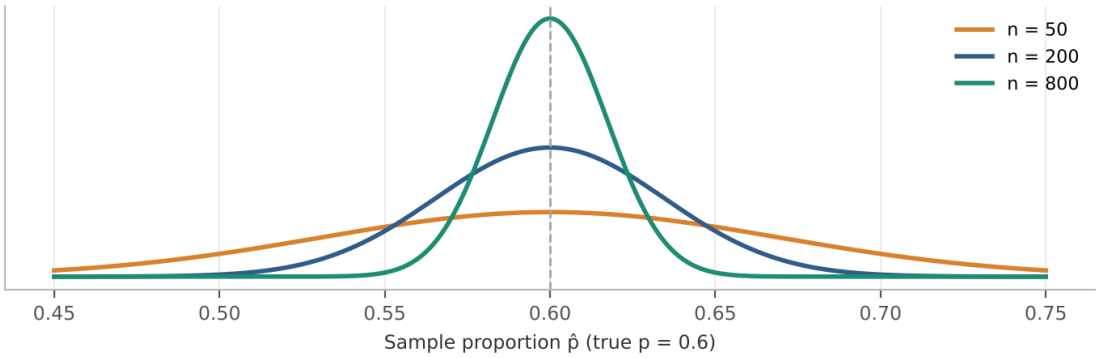


Figure 2. Sampling distribution of $\hat{p}$ for three poll sizes. Bigger polls give tighter estimates around the true p.

6. Other statistics also have sampling distributions

Statistic	Sampling distribution (under standard conditions)
Sample mean $\bar{x}$, σ known	Normal (exactly if the population is normal, approximately otherwise)
Standardized mean, σ unknown, using s	Student's t with n − 1 degrees of freedom (Topic 09)
Sample proportion $\hat{p}$	Approximately normal when np and n(1 − p) are at least 10
Sample variance s^2 (normal population)	$(n-1)s^2/\sigma^2$ follows a chi-square distribution with n − 1 df
Difference of two means	Approximately normal, with the standard errors combined (Topic 07)

7. Finite populations

The formula σ/√n assumes we sample from a very large (or infinite) population, or with replacement. If we sample **more than about 5 to 10%** of a small population without replacement, the sample captures a big share of the population and $\bar{x}$ is more precise than σ/√n suggests. A correction factor fixes this:

$$\mathrm{SE} = \frac{\sigma}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}$$

WORKED EXAMPLE **Small class**

A class of N = 120 students is sampled with n = 40 and σ = 15. Uncorrected SE = 15/√40 = 2.372. The correction factor is √(80/119) = 0.820, giving SE = **1.945**. If instead N were 10,000, the factor would be 0.9980, essentially 1.

8. Common mistakes

- Confusing the standard deviation of the **data** (s or σ) with the standard deviation of the **sample mean** (σ/√n).
- Thinking a larger sample changes the shape or the spread of the **population**. It only changes the sampling distribution.
- Believing $\bar{x}$ equals μ. It is a random value near μ, and how near is exactly what the sampling distribution measures.
- Using σ/√n when the sample is a large fraction of a small population and ignoring the finite population correction.
- Treating the 'independence' of observations lightly. Clustered or repeated measures break the σ/√n rule.

9. Simulate it yourself

```
import numpy as np
rng = np.random.default_rng(1)

population = rng.exponential(scale=50, size=100_000) + 10      # skewed population
mu, sigma = population.mean(), population.std()

n = 30
sample_means = np.array([rng.choice(population, n).mean() for _ in range(5000)])

print(sample_means.mean(), mu)                 # centre matches mu
print(sample_means.std(), sigma/np.sqrt(n))    # spread matches sigma / sqrt(n)
```

SUMMARY

- A statistic varies from sample to sample, so it has a **sampling distribution**.
- $E(\bar{x}) = \mu$ and $SD(\bar{x}) = \sigma/\sqrt{n}$; for a normal population, $\bar{x}$ is exactly normal.
- $\hat{p}$ has mean p and SD $\sqrt{p(1-p)/n}$, and is roughly normal if $np \geq 10$ and $n(1-p) \geq 10$.
- Averages are less variable than individual values; quadrupling n halves the spread.
- Population, sample and sampling distributions are three different things.

Practice questions

Try each question before reading the answer underneath it.

Q1. A population has $\mu = 80$ and $\sigma = 12$. For $n = 36$, what are the mean and standard deviation of the sampling distribution of $\bar{x}$?

Answer. Mean = 80; SD = $12/\sqrt{36} = 2$.

Q2. In Q1 the population is normal. Find $P(\bar{x} < 77)$.

Answer. $z = (77 - 80)/2 = -1.5$, probability 0.0668.

Q3. Is $\hat{p}$ approximately normal if $p = 0.03$ and $n = 100$?

Answer. No: $np = 3$ is below 10, so the success-failure condition fails.

Q4. $p = 0.4$ and $n = 150$. Find $SD(\hat{p})$.

Answer. $\sqrt{(0.4 \times 0.6/150)} = 0.0400$.

Q5. By what factor must n increase to cut the standard deviation of $\bar{x}$ in half?

Answer. By a factor of 4, because $SD(\bar{x}) = \sigma/\sqrt{n}$.

Q6. Is the sampling distribution of $\bar{x}$ the same as the distribution of the data in one sample?

Answer. No. The data distribution describes individual values (spread s); the sampling distribution describes the $\bar{x}$ from many samples (spread $\sigma/\sqrt{n}$).